How-To Library

Informatica

Deploying PowerCenter on AWS ICMP Infrastructure

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Abstract

You can deploy a PowerCenter® installation on AWS ICMP. This article describes how to verify that the PowerCenter installation is correctly deployed in your AWS ICMP infrastructure.

Supported Versions

- PowerCenter 10.1.1

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PowerCenter on AWS ICMP Overview

PowerCenter is a data integration product that transforms fragmented, raw data from any source, at any latency into complete, actionable information. You can integrate data from on-premises data sources, cloud data sources, or Amazon Web Services (AWS) with PowerCenter.

With PowerCenter on AWS, you can deliver business value quickly. You can run the exact same data integration environment in the AWS cloud or on-premises seamlessly according to your requirements with the same experience. You can update AWS infrastructure resources based on your needs. You can work with mappings, workflows, and metadata across on-premises and AWS cloud environments.

To ensure that PowerCenter works on AWS seamlessly, select the right AWS resources to optimize the PowerCenter environment. You can use an Amazon EC2 instance with any supported operating system for the migration of PowerCenter applications.

Intelligence Community Marketplace (ICMP) is an AWS Marketplace for the United States Intelligence Community (IC) that caters to the needs of the IC customers. With ICMP, you can easily discover, purchase, and deploy software packages and applications from vendors with a strong presence in the IC in a cloud that is not connected to the public internet. For more information about the United States Intelligence Community Marketplace, contact AWS.

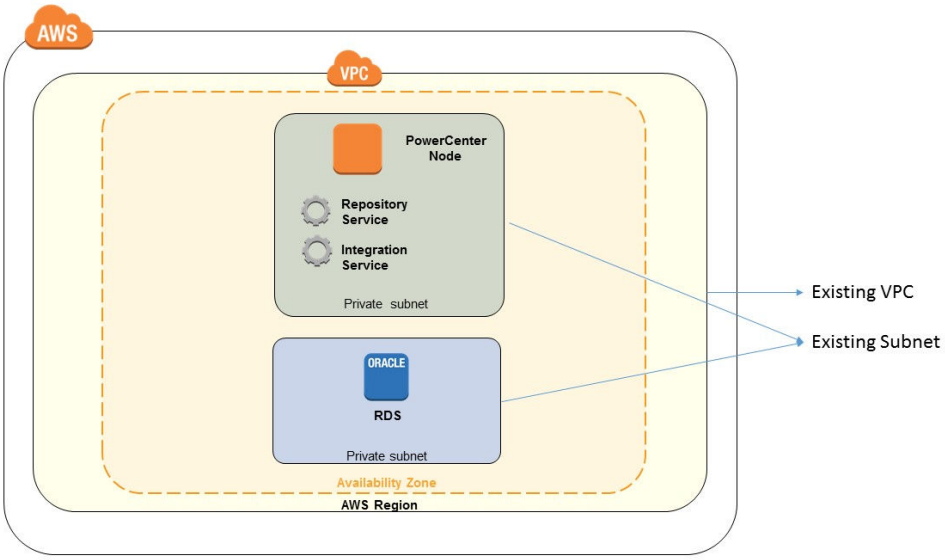
This article describes how you can ensure that the PowerCenter installation is correctly deployed in your AWS ICMP infrastructure.

PowerCenter on AWS ICMP Offering Details

You can deploy Informatica PowerCenter version 10.1.1 with a single node configuration in an existing VPC infrastructure on Windows Server 2012 R2 or Red Hat Enterprise Linux 7.3 along with Oracle RDS (12.1 and 11.2) as BYOL for the domain database. If you create a stack from the Marketplace listing, you will be billed for the AWS resources used.

PowerCenter on AWS ICMP Architecture Diagram

The following diagram shows the PowerCenter on AWS ICMP architecture using an AWS CloudFormation Script (CFS):



Steps to Check the Deployment Status of AWS Stack and Informatica Installation

1. Check the stack creation.

   Check that the stack creation is successful. Stack creation failure can be linked to user access, incomplete creation of the Oracle RDS instance, or the node instance. Review the stack log to troubleshoot the issue and restart the stack creation again. If the node stack creation fails, you can also check on the zone availability for the required instances.

2. Check the Informatica domain.

   a. Check if you can login to the EC2 instance.

   b. If the domain runs on the Windows EC2 instance, review the installation log in the following path:

      `C:\Informatica\10.1.1\Informatica_10.1.1_Services_<time_stamp>.log`

      If the domain runs on Linux, review the installation log in the following path:

      `/home/ec2-user/Informatica/10.1.1`

   c. Ensure that the domain and all the service processes are running and that the Administrator tool is accessible.

   d. Check if any security issues exist, such as ports getting blocked.

3. Check the Informatica services.
   a. If the domain is available but the related PowerCenter Repository Service and Integration Service are inaccessible for Windows, check the custom logs in the following location:

      `C:\ec2-user\InfaServiceLog.log`

      If the domain runs on Linux, review the custom logs in the following location:

      `/home/ec2-user/InfaServiceLog.log`

   b. If the issue persists and the domain runs on Windows, review the service logs in the following location:

      `C:\Informatica\10.1.1\logs\<nodename>\<Servicename>`

      If the domain runs on Linux, review the custom logs in the following location:

      `/home/ec2-user/Informatica/10.1.1/logs/<nodename>/<ServiceName>`
4. Check the PowerCenter deployment

   Verify the Administrator tool access and the PowerCenter configuration.

Check the PowerCenter Deployment

Verify that you can access the Administrator tool and the PowerCenter configuration.

Verify Access to the Administrator Tool

To verify the success of the PowerCenter deployment, review the output in the AWS ICMP Console (Stack details). Or, create an Administrator tool instance.

1. In the AWS ICMP Console (Stack details), select the master gateway EC2 instance.
2. From the description pane, copy the Private IP or DNS.
3. Verify that you can access the EC2 instance from your machine. To access the Administrator tool in the browser, enter the Private IP or DNS followed by the Administrator tool port number.

   Use the following format:

   `http://<Private IP or DNS>:<Administrator_tool_port>`

   Use the default Administrator tool port number of `6008`.

   Or, select the Administrator tool link in the output section of the created stack.
4. Verify that you can view the Administrator tool login page.
5. Log in to the Administrator tool.
6. Verify that the default PowerCenter Integration Service and the PowerCenter Repository Service are enabled.

Verify the PowerCenter Configuration

After you have successfully deployed PowerCenter from the Marketplace listing, you can test that the PowerCenter Repository Service and Integration Service work effectively with an import of sample files using the command line.

Step 1. Configure Environment Variables

Configure the environment variables to work with the Informatica installation.

The following table lists the environment variables that you need to set on UNIX:

Variable	Description
INFA_HOME	Use INFA_HOME to designate the Informatica installation directory. If you modify the Informatica directory structure, you need to set the environment variable to the location of the Informatica installation directory or the directory where the installed Informatica files are located. For example, you use a softlink in UNIX for any of the Informatica directories. To configure INFA_HOME so that any Informatica application or service can locate the other Informatica components it needs to run, set INFA_HOME to the location of the Informatica installation directory. After you set the environment variable, enter the export command in the command line in the following format: `export INFA_HOME=/home/ec2-user/Informatica/10.1.1`
JRE_HOME	Configure the library path environment variable to point to the installed Java directory on machines where the PowerCenter Integration Service process runs. After you set the environment variable, enter the export command in the command line in the following format: `export JRE_HOME=/home/ec2-user/Informatica/10.1.1/java/jre`
JAVA_HOME	Configure the library path environment variable to point to the installed Java directory on machines where the PowerCenter Integration Service process runs. After you set the environment variable, enter the export command in the command line in the following format: `export JAVA_HOME=/home/ec2-user/Informatica/10.1.1/java/jre`
ORACLE_HOME	Set the variable to the Oracle client installation directory. After you set the environment variable, enter the export command in the command line in the following format: `export ORACLE_HOME=/home/oracle/product/11.2.0.4`
LD_LIBRARY_PATH	Configure the library path environment variable on the machine that run the Integration Service and Repository Service processes. The shared library path must also include the Informatica installation directory (server_dir). After you set the environment variable, enter the export command in the command line in the following format: `export LD_LIBRARY_PATH=/home/ec2-user/Informatica/10.1.1/server/bin:/home/ec2-user/Informatica/10.1.1/java/jre/lib/amd64/server::/home/oracle/product/11.2.0.4/lib:/home/oracle/product/11.2.0.4/lib:/bin:`

Variable	Description
PATH	The installer appends file paths required by Informatica to the PATH environment variable. Verify that the length of the PATH environment variable does not exceed the system limits. Verify that the PATH environment variable includes the Oracle bin directory. After you set the environment variable, enter the export command in the command line in the following format: `export PATH=/home/ec2-user/Informatica/10.1.1/server/bin:/bin:/usr/xpg4/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/bin:/usr/bin:/sbin:/usr/sbin:/usr/local/bin`
TNS_ADMIN	If the tnsnames.ora file is not in the same location as the Oracle client installation location, set the TNS_ADMIN environment variable to the directory where the tnsnames.ora file resides. After you set the environment variable, enter the export command in the command line in the following format: `export TNS_ADMIN=/home/oracle/product/11.2.0.4/network/admin`

Step 2. Connect to the Repository and Create a Folder

1. To connect to the repository, enter the `pmrep connect` command in the command line on UNIX.

   For example, enter the following command:

   ```
   $ /home/ec2-user/Informatica/10.1.1/server/bin/pmrep connect -r PowerCenterRepository -d InfaDomain -n Administrator -x Administrator@123
   ```

   When the connection is successful, the following message appears in the command line:

   ```
   Connected to repository <repository_name> in <domain_name> as user <user_name>
   connect completed successfully.
   ```

2. To create a folder, enter the `pmrep createfolder` command in the command line.

   For example, enter the following command:

   ```
   $ pmrep createfolder -n PCTest
   ```

   When the folder gets successfully created, the status message appears in the command line in the following format:

   ```
   Creating folder <folder_name> in repository <repository_name>
   createfolder completed successfully.
   ```

Step 3. Import Workflow with Sample Files

You can import a workflow with sample files for workflow export and control file to create metadata in the repository.

1. Save the sample workflow export file, `wf_mp_pc_test.xml` and control file, `ctrl.xml` on your system and copy the files to the user home location using FTP.

   You can download the zip file containing these files from the following location:

   https://kb.informatica.com/h2l/HowTo%20Library/1/1108-PConAWSICMP-Sample.zip

2. In the command line, to import the workflow, enter the `pmrep objectimport` command.

   For example, enter the following command:

   ```
   $ /home/ec2-user/Informatica/10.1.1/server/bin/pmrep objectimport -i /home/ec2-user/wf_mp_pc_test.xml -c /home/ec2-user/ctrl.xml
   ```

When all the objects are successfully imported, the following message appears in the command line:

```
workflow <workflow_name> tasks validation completed with no errors.
Workflow <workflow_name> inserted.
objectimport completed successfully
```

Step 4. Enter the Sample Source File in the Source File Location

Insert the source file in the source file location on the machine where the PowerCenter Integration Service runs.

1. Navigate to the source file location on the machine where the PowerCenter Integration Service runs.

   By default, the source files reside in the following location:

   `$INFA_HOME/server/infa_shared/SrcFiles`

   $INFA_HOME is the location where Informatica product is installed.

2. Copy the sample, `FF_Src` file into the source file location using FTP.

   You can download the zip file containing the source file from the following location:

   https://kb.informatica.com/h2l/HowTo%20Library/1/1108-PConAWSICMP-Sample.zip

Step 5. Run the Workflow

1. To start the workflow, enter the `pmcmd startworkflow` command in the command line.

   For example, enter the following command:

   ```
   $ /home/ec2-user/Informatica/10.1.1/server/bin/pmcmd startworkflow -d InfaDomain -sv
   PowerCenterIntegrationService -u Administrator -p Administrator@123 -f PCTest -wait
   wf_mp_pc_test
   ```

   When the workflow starts, you can see the progress of the workflow run in the command line.

   It will show succeeded once the run is complete in the following format: `INFO: Workflow [workflow_name]: Execution succeeded.`

2. To test the workflow run status, enter the `pmcmd getworkflowdetails` command in the command line.

   For example, enter the following command:

   ```
   $ /home/ec2-user/Informatica/10.1.1/server/bin/pmcmd getworkflowdetails -d InfaDomain -sv
   PowerCenterIntegrationService -u Administrator -p Administrator@123 -f PCTest wf_mp_pc_test
   ```

   Ensure that the workflow run status shows as succeeded. A successful message implies that the PowerCenter services are configured and works as expected.

3. You can also view the target file in the following location: `$INFA_HOME/server/infa_shared/TgtFiles`

   For example, enter the following command:

   ```
   $ more /home/ec2-user/Informatica/10.1.1/server/infa_shared/TgtFiles/FF_Tgt.out
   ```

   In the sample FF_Tgt.out file, verify if the contents match that of the source file as follows:

   ```
   abc,def
   dfs,ghj
   ```

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